

Creating static/dynamic link library

<https://www.geeksforgeeks.org/static-vs-dynamic-libraries/>

Static library:

- Creation
 - a. Create a C file that contains functions in your library (Example: lib_mylib.c)
 - b. Create a header file for the library

```
/* Filename: lib_mylib.h */  
void fun(void);
```
 - c. Compile library files: gcc -c lib_mylib.c -o lib_mylib.o
 - d. Create static library: ar rcs lib_mylib.a lib_mylib.o
- Usage
 - a. In the C file include the library (Example: #include "lib_mylib.h")
 - b. Compile the driver program (Example: gcc -c driver.c -o driver.o)
 - c. Link the compiled driver program to the static library: gcc -o driver driver.o -L. -l_mylib
 - d. Run the driver program

Dynamic library:

https://medium.com/@The_Mad_Zaafa/creating-and-using-dynamic-libraries-c-a9d344822ed0

<https://www.youtube.com/watch?v=UGm0ze1kmE8>

- Creation
 - a. Compiling the library C file: gcc lib_mylib.c -c -fPIC -o lib_mylib.o
 - b. Create lib_mylib.h file, including the declarations of the functions
 - c. gcc -shared -o lib_mylib.so lib_mylib.o
 - d. export the path for libraries: export
LD_LIBRARY_PATH=\$PWD:\$LD_LIBRARY_PATH
- Usage
 - a. In the driver program include : #include "lib_mylib.h"
 - b. Compile the driver program using the following command: gcc -L. -o test_code test_code.c -l_mylib